

NORTH SYDNEY BOYS HIGH SCHOOL

2015 YEAR 11 ASSESSMENT TASK 2

Mathematics Extension 1

General Instructions

- Working time – 50 minutes
- **Reading time – 5 minutes**
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher: Please colour

- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Mr Weiss
- ☐ Mr Ziariaris

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	5	6	Total	Total
Mark	$\frac{8}{8}$	$\frac{8}{8}$	$\frac{8}{8}$	$\frac{8}{8}$	$\frac{8}{8}$	$\frac{6}{6}$	$\frac{46}{46}$	$\frac{100}{100}$

YEAR 11 ASSESSMENT TASK 2 2015

QUESTION 1 START A NEW PAGE

- (a) Differentiate
- (i) $(5x^2 - 4)^5$ 1
- (ii) $\frac{2x^3 + 3x - 4}{x}$ 2
- (iii) $2x\sqrt{x}$ 2
- (b) Find the equation of the normal to the curve $y = 5x^2 - 3x + 1$ at the point where $x = -1$. 3

QUESTION 2 START A NEW PAGE

- (a) Find the equation of the straight line which passes through the point C(4, -1) and is parallel to the line through A(-1, 2) and B(2,7). 2
- (b) Graph the intersection of the regions
 $y < 3x$ and $x - 3y + 1 \leq 0$ 3
- (c) The point C (6, 2) divides the interval AB in the ratio 2:3. If A has coordinates (4, -2) find the coordinates of B. 3

QUESTION 3 START A NEW PAGE

(a) Express 140° in radians as an exact value. 1

(b) Given that $\operatorname{cosec} \alpha = 4$ where α is acute,
Find the value of $\cos 2\alpha$. 3

(c) Find the exact value of $\frac{1}{4} \sin \frac{\pi}{8} \cos \frac{\pi}{8}$ 2

(d) Solve for $0^\circ \leq \theta \leq 360^\circ$

$$\cos \frac{\theta}{2} = \frac{1}{2} \quad 2$$

QUESTION 4 START A NEW PAGE

(a) For what values of x is $f(x) = 3x^3 - 2x^2 - 5x - 4$ increasing. 3

(b) Find $\lim_{x \rightarrow 2} \frac{2x^2 - 8}{x - 2}$ 2

(c) A function is defined by

$$f(x) = \begin{cases} ax^2 + b & \text{for } x < 0 \\ ax^3 - 6 & \text{for } x > 0 \end{cases}$$

Given that $f'(1) - f'(-1) = -15$, show that $f(2) = -30$ 3

exam continues overleaf.....

QUESTION 5 START A NEW PAGE

- (a) Find the acute angle between the lines $5x - 6y = 30$
and $4x + 3y = 12$ giving your answer to the nearest minute. 3
- (b) Find the perpendicular distance between the parallel lines
 $3x + 4y = -1$ and $3x + 4y = 4$. 2
- (c) Solve for $-180^\circ \leq \theta \leq 270^\circ$
 $2 \cos(\theta - 225)^\circ = \sqrt{3}$ 3

QUESTION 6 START A NEW PAGE

- (i) Show that $\left(\frac{\sin 3x}{\sin x}\right)^2 - \left(\frac{\cos 3x}{\cos x}\right)^2 = 8 \cos 2x$
for $\sin x \neq 0$ and $\cos x \neq 0$ 3
- (ii) Hence or otherwise solve
 $\left(\frac{\sin 3x}{\sin x}\right)^2 - \left(\frac{\cos 3x}{\cos x}\right)^2 = 8 \sin x$ where $0^\circ \leq \theta \leq 360^\circ$ 3

END OF EXAMINATION

ASSESS TASK 2 YR 11 2015.

1. (a) (i) $5(5x^2-4)^4 \cdot 10x$
 $= 50x(5x^2-4)^4$

(ii) $\frac{2x^3}{x} + \frac{3x}{x} - \frac{4}{x}$
 $= 2x^2 + 3 - 4x^{-1}$
 $\frac{dy}{dx} = 4x + \frac{4}{x^2}$

(iii) $2x^{3/2}$
 $\frac{dy}{dx} = 3x^{1/2}$

(b) $y = 5x^2 - 3x + 1$
 $\frac{dy}{dx} = 10x - 3$

At $x = -1$

$\frac{dy}{dx} = -13$

∴ m of normal = $\frac{1}{13}$ at $(-1, 9)$

Equn of normal

$y - 9 = \frac{1}{13}(x + 1)$

$13y - 117 = x + 1$

$x - 13y + 118 = 0$

2. (a) $m = \frac{7-2}{2+1}$

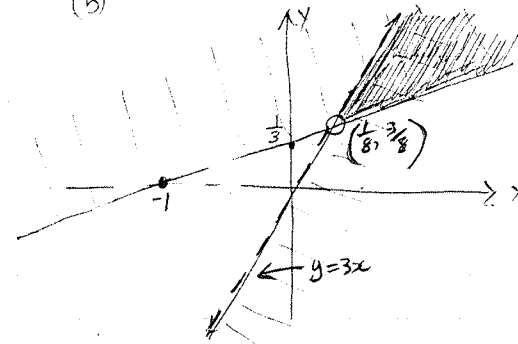
$m = \frac{5}{3}$

Eqn: $y + 1 = \frac{5}{3}(x - 4)$

$3y + 3 = 5x - 20$

$5x - 3y - 23 = 0$

(b)



c) $(6, 2) = \frac{nx_1 + mx_2}{m+n} \quad \frac{ny_1 + my_2}{m+n}$

$(6, 2) = \frac{3(4) + 2(x_2)}{5}, \frac{3(-2) + 2y_2}{5}$

∴ $6 = \frac{12 + 2x_2}{5}, \quad 2 = \frac{-6 + 2y_2}{5}$

$\frac{30 - 12}{2} = x_2$

$x_2 = 9$

$10 = -6 + 2y_2$

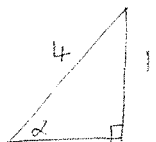
$y_2 = 8$

∴ B(9, 8)

OR you could use diagram form

$$3. a) \quad 140 \times \frac{\pi}{180} = \frac{7\pi}{9}$$

$$b) \quad \sin \alpha = \frac{1}{4}$$



$$\begin{aligned} \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha \\ &= 1 - 2\sin^2 \alpha \\ &= 1 - 2\left(\frac{1}{4}\right)^2 \\ &= 1 - \frac{1}{8} \\ &= \frac{7}{8} \end{aligned}$$

$$\begin{aligned} c) \quad \frac{1}{4} \sin \frac{\pi}{8} \cos \frac{\pi}{8} \\ &= \frac{1}{4} \times \frac{1}{2} \sin 2\left(\frac{\pi}{8}\right) \\ &= \frac{1}{8} \sin \frac{\pi}{4} \\ &= \frac{1}{8} \times \frac{1}{\sqrt{2}} \\ &= \frac{1}{8\sqrt{2}} \\ &= \frac{\sqrt{2}}{16} \end{aligned}$$

$$\begin{aligned} d) \quad \cos \frac{\theta}{2} &= \frac{1}{2} \\ \frac{\theta}{2} &= 60^\circ \\ \theta &= 120^\circ \end{aligned}$$

$$0^\circ \leq \frac{\theta}{2} \leq 180^\circ$$

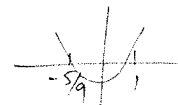
$$\begin{aligned} 4. a) \quad f(x) &= 3x^3 - 2x^2 - 5x - 4 \\ f'(x) &= 9x^2 - 4x - 5 \end{aligned}$$

Inc. when $f'(x) > 0$

$$9x^2 - 4x - 5 > 0$$

$$\begin{array}{r} 9x \times 5 \\ x - 1 \end{array}$$

$$(9x + 5)(x - 1) > 0$$



Inc when $x < -\frac{5}{9}$ OR $x > 1$

$$\begin{aligned} b) \quad \lim_{x \rightarrow 2} \frac{2(x^2 - 4)}{x - 2} \\ &= \lim_{x \rightarrow 2} 2(x + 2) \\ &= 8 \end{aligned}$$

$$\begin{aligned} c) \quad f(2) &= 8a - 6 \\ f'(x) &= 3ax^2 \quad x > 0 \\ f'(1) &= 3a \\ f'(x) &= 2ax \quad x < 0 \\ f'(-1) &= -2a \end{aligned}$$

$$\therefore 3a - 2a = -15$$

$$5a = -15$$

$$a = -3$$

$$\text{Now } f(2) = 8a - 6$$

$$= -24 - 6$$

$$= -30$$

$$5. a) m_1 = \frac{5}{6}$$

$$m_2 = -\frac{4}{3}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{\frac{5}{6} + \frac{4}{3}}{1 - \frac{5}{6} \times \frac{4}{3}} \right|$$

$$= \left| \frac{\frac{13}{6}}{1 - \frac{20}{18}} \right|$$

$$= \left| \frac{13}{6} \div -\frac{1}{9} \right|$$

$$= \left| \frac{13}{6} \times \frac{9}{-1} \right|$$

$$= \left| -\frac{39}{2} \right|$$

$$\theta = \tan^{-1} \frac{39}{2}$$

$$\theta = 87^\circ 4'$$

b) Point on Line $3x + 4y = 4$

$$\begin{matrix} a & b & c & (x_1, y_1) \\ 3x + 4y & +1 = 0 & (0, 1) \end{matrix}$$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \left| \frac{0 + 4 + 1}{\sqrt{9 + 16}} \right|$$

$$= \left| \frac{5}{5} \right|$$

$$= 1 \text{ unit.}$$

$$-180^\circ \leq \theta \leq 270^\circ$$

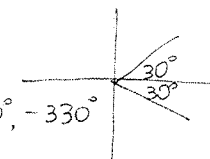
$$c) \cos(\theta - 225^\circ) = \frac{\sqrt{3}}{2}$$

$$-405^\circ \leq \theta - 225^\circ \leq 45^\circ$$

$$(\theta - 225^\circ)_b = 30^\circ$$

Consider 1st + 4th Quads.

$$\theta - 225^\circ = 30^\circ, -30^\circ, -390^\circ, -330^\circ$$



$$\therefore \theta = 255^\circ, 195^\circ, +165^\circ, -105^\circ$$

$$\begin{aligned}
 6. \quad \text{In LHS} &= \left(\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x} \right) \left(\frac{\sin 3x}{\sin x} + \frac{\cos 3x}{\cos x} \right) \\
 &= \left(\frac{\cos x \sin 3x - \cos 3x \sin x}{\sin x \cos x} \right) \left(\frac{\cos x \sin 3x + \cos 3x \sin x}{\sin x \cos x} \right) \\
 &= \left(\frac{\sin(3x-x)}{\frac{1}{2} \sin 2x} \right) \left(\frac{\sin(3x+x)}{\frac{1}{2} \sin 2x} \right) \\
 &= \left(\frac{\sin 2x}{\frac{1}{2} \sin 2x} \right) \frac{2 \sin 4x}{\sin 2x} \\
 &= \frac{4 \sin 4x}{\sin 2x} \\
 &= \frac{4 \cdot 2 \sin 2x \cos 2x}{\sin 2x} \\
 &= 8 \cos 2x \\
 &= \text{RHS.}
 \end{aligned}$$

$$(ii) \quad 8 \cos 2x = 8 \sin x$$

$$\cos 2x = \sin x$$

$$1 - 2 \sin^2 x = \sin x$$

$$2 \sin^2 x + \sin x - 1 = 0$$

$$\frac{2 \sin x}{\sin x} \times$$

$$(2 \sin x - 1)(\sin x + 1) = 0$$

$$\sin x = \frac{1}{2}, \sin x = -1$$

$$x = 30^\circ, 150^\circ, 270^\circ$$

But $x \neq 270$ as $\cos x$ in denom.

∴ Only soln $x = 30^\circ, 150^\circ$